

RAAHUL MUTHUKRISHNAN

Research Sciences – Computer Vision, Multimodal Machine Learning & Machine Intelligence

Mumbai, India | +91 8928267607 | raahul4004@gmail.com | linkedin.com/in/raahul404 | github.com/raahul4004

RESEARCH SUMMARY

Master's student in Artificial Intelligence & Machine Learning with applied research experience at Amazon, currently first author on a manuscript in preparation. Research interests span computer vision, multimodal and vision-language learning, and model robustness under distribution shift. Works the full research cycle: reproducing state-of-the-art papers, forming a hypothesis, designing controlled experiments on large-scale datasets, and implementing the result as a scalable system. Strong in Python and PyTorch, having written transformers and inference engines from scratch; also works in C/C++.

EDUCATION

Vellore Institute of Technology

Jul 2025 – May 2028

MS (Master of Science), Artificial Intelligence & Machine Learning GPA: 9.63/10.0

Ramniranjan Anandilal Podar College

Aug 2022 – May 2025

BSc (Bachelor of Science), Data Science & Analytics GPA: 10.0/10.0

RESEARCH EXPERIENCE

Applied Scientist Intern

May 2026 – Present

Amazon – VISTA (Computer Vision, Multimodal Learning, Content Authenticity)

Bangalore, India

- Leading an independent research thread on cross-generator generalization in AI-generated image detection: surveyed the literature, found that reported accuracy did not transfer to unseen image sources, and designed the study to isolate why.
- Proposed and trained **SRC-Net**, adapting a frozen **CLIP** vision-language backbone by **parameter-efficient fine-tuning** — **0.965 average precision** on unseen sources with only **16.6M** trainable parameters.
- Found that **confounded preprocessing** (JPEG compression, crop, resolution), not model capability, accounted for much of the accuracy reported by prior work; built controlled benchmarks that separate the two and re-evaluated the field's baselines against them.
- Reproduced and benchmarked **10+ state-of-the-art architectures** from their original papers on large-scale image datasets (accuracy, latency, throughput, memory), grounding model selection in measurement rather than reported numbers.
- Collaborated with researchers and product partners to take the prototype to a production-level system at **27.7 ms/image** (~2× faster than every baseline); presented findings internally and is preparing the work for conference submission.

PUBLICATIONS

R. Muthukrishnan et al. “Cross-Generator Generalization in AI-Generated Image Detection.” *Lead author, manuscript in preparation for conference submission, 2026.*

RESEARCH PROJECTS

Large Language Model Inference Engine from Scratch | *Python, PyTorch*

[GitHub](#)

- Implemented a full **autoregressive transformer decoder** (multi-head self-attention, feed-forward blocks, rotary positional embeddings) from the original papers with no reference implementation.
- Improved algorithm performance with **KV-caching**, cutting latency **65%** and memory **40%**; built dual CPU/GPU execution paths.

Retrieval-Augmented Generation for Document QA | *Python, PyTorch, Hugging Face*

[GitHub](#)

- Built a **scalable retrieval system** over 500+ page corpora pairing dense-embedding semantic search with a generative model, applying **natural language understanding** to ground answers in evidence.
- Ran systematic ablations over chunking and embedding strategies, cutting retrieval latency from **2.3s to 180ms** at 94% relevance; validated on **200+ held-out queries**.

Spatio-Temporal Video Recognition with 3D Convolutional Networks | *PyTorch Lightning, OpenCV*

[GitHub](#)

- Trained a **3D convolutional network** modeling motion and frame-to-frame coherence for real-time video classification — **92% test accuracy** across varied lighting, angle, and occlusion; temporal augmentation improved out-of-distribution robustness **8%**.

HONORS & COMPETITIONS

Kaggle NeuroGolf Championship (IJCAI-ECAI 2026) — Top 24%, \$50K model-compression track

Amazon ML Challenge — All India Rank 38 of 10,000+ teams

TECHNICAL SKILLS

Research Areas: computer vision, natural language understanding, multimodal / vision-language models, machine intelligence, deep learning, data mining, optimization

Programming & Frameworks: Python, C/C++, SQL, PyTorch, PyTorch Lightning, Hugging Face Transformers, Scikit-learn, OpenCV, NumPy, Pandas, Git, Linux, cloud GPU training

Modeling & Experimentation: transformers, 3D convolutional networks, self-supervised learning, parameter-efficient fine-tuning (PEFT / LoRA), distillation, quantization, experimental design, large-scale data processing, benchmark construction, ablation studies, robustness analysis